SOAL 7

Permasalahan:

Z = 35,25

X1 = 3,75

X2 = 1,5

JAWABAN

Iterasi 1

Subset 1

Maximize Z = 7X1 + 6X2

Subject to

$2X1 + 3X2 \leq 12$

$6X1 + 5X2 \leq 30$

$\mathbf{X2 \geq 2}$

Perhitungan subset 1

- **Mencari X1**

  2X1 + 3X2 = 12

  2X1 + 3(2) = 12

  2X1 + 6 = 12

  2X1 = 6

  X1 = 3

  6X1 + 5X2 = 30

  6X1 + 5(2) = 30

  6X1 + 10 = 30

  6X1 = 20

  $X1 = \frac{20}{6} = 3{,}33$

- **Mencari Z**

  Z = 7X1 + 6X2

  Z = 7(3) + 6(2)

  Z = 33 **(Calon Solusi)**

  Z = 7X1 + 6X2

  $Z = 7(\frac{20}{6}) + 6(2)$

  Z = 35,31 → **Solusi tidak digunakan** karena melebihi batas Z pada permasalahan

Subset 2

Maximize Z = 7X1 + 6X2

Subject to

$2X1 + 3X2 \leq 12$

$6X1 + 5X2 \leq 30$

$\mathbf{X2 \leq 1}$

Perhitungan subset 2

- **Mencari X1**

  2X1 + 3X2 = 12

  2X1 + 3(1) = 12

  2X1 + 3 = 12

  2X1 = 9

  X1 = 4,5

  6X1 + 5X2 = 30

  6X1 + 6(1) = 30

  6X1 + 6 = 30

  6X1 = 24

  X1 = 4

- **Mencari Z**

  Z = 7X1 + 6X2

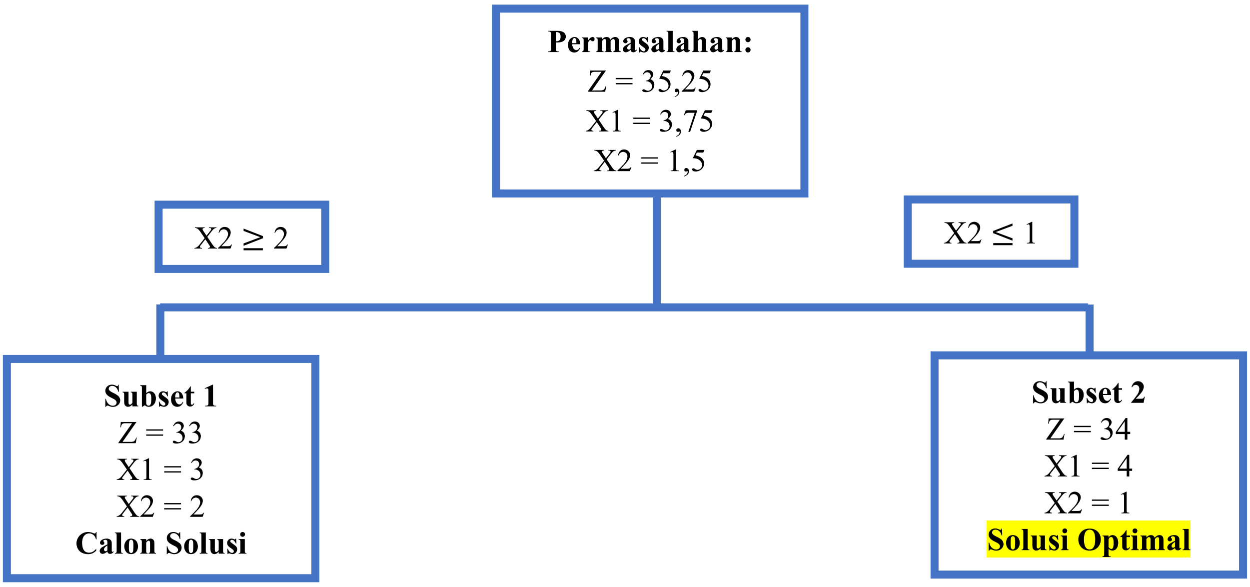
  Z = 7(4,5) + 6(1)

  Z = 37,5 → **Solusi tidak digunakan** karena elebihi batas Z pada permasalahan

  Z = 7X1 + 6X2

  Z = 7(4) + 6(1)

  Z = 34 **(Solusi Optimal)**


Permasalahan:
Z = 35,25
X1 = 3,75
X2 = 1,5
X2 ≥ 2
X2 ≤ 1
Subset 1
Z = 33
X1 = 3
X2 = 2
Calon Solusi
Subset 2
Z = 34
X1 = 4
X2 = 1
Solusi Optimal